1. Quadratic models.

① Def: A quadratic model over k is a semifree extension $A = k[X]$ where $|X_i|$ is finite and:

$$\partial(X) \subset kX^2$$

↓ means only quadratic terms

Then ∂ raises the inherent degree:

$$\partial(X^n) \subset kX^{n+1}.$$

Rmk: For $J = (X_{\geq 0})$ in $\mathbb{Q}[X]$,

$$\partial(J) \subset J^2 \quad \text{and} \quad \partial(J^n) \subset J^{n+1}.$$

② Def: Let $k[X]$ be a minimal semifree extn. of k, then the quadratic part of $k[X]$ is:

$$quad(k[X]) = k[X : \partial_q(X) = \partial_{[2]}(X)].$$

where $\partial = \partial_{[2]} + \partial_{[3]} + \cdots$, $\partial_{[2]}$ means the part in kX^2.

Rmk: ∂_q is a well-def. differential since for minimal $k[X]$, terms like $\partial_{[2]}\partial_{[3]}$ must have larger word length, so $\partial^2 = 0 \Rightarrow \partial_q^2 = 0$.

③ Quadratic part of a minimal model.

Note that for a minimal model $A := \mathbb{Q}[X]$, its quadratic part

means $\mathrm{grad}(k[X])$.

- Consider the associated graded ring of $Q[X]$ w.r.t. $(X) = (X_{\geq 0})$:

$$gr_{(X)}(Q[X]) = \bigoplus_{n\geq 0} X^n/X^{n+1} \cong k \oplus kX^1 \oplus kX^2 \oplus \cdots$$

since: $\begin{cases} X^0 = Q[X], \text{ and } Q[X]/(m + X_{\geq 1}) \cong k \\ \text{Similarly, } X^n/X^{n+1} \text{ consists of length } n \text{ variables with coeff. in } k. \end{cases}$

- Differential ∂' on $gr_{(X)}(Q[X])$ is induced by ∂ on $Q[X]$ since $\partial(X^n) \subset \partial(X^{n+1})$.
  $\partial' = \partial_Q$ since ∂ sends $x \in kX$ only to quadratic terms in kX^2:

$$[\partial'(x)] \equiv [\partial(x)] \pmod{X^3} \quad \text{by definition.}$$

④ Def: A <u>map of quadratic model</u> $\varphi: k[X] \to k[Y]$ is map of dgas such that $\varphi(X) \subset kY$.

Def: $(-)^\vee := \mathrm{Hom}_k(-,k)$, and $(kX)^\vee = \prod_{x\in X} k$ since kX means v.s. freely gen. by elts of X.

2. Homotopy Lie algebra.

① Unravelling the definitions.

Def: Let $k[X]$ be a quadratic semifree dga. The <u>homotopy Lie algera</u> of $A := k[X]$ is:

$$\pi^*(A) = (\underset{\downarrow}{\Sigma} kX)^\vee$$

shifted since $[-,-]$ must be deg. 0.

such that:

$$[u,v](x) := \langle u\otimes v - (-1)^{|u||v|} v\otimes u,\ \partial(x)\rangle, \qquad (\text{and } w^{[2]}\dots).$$

$[u,v](x)$ means natural pairing $\langle [u,v], x\rangle$

$\partial(x)$: sum of quadratic terms

RHS is: if $\partial(x) = yz$,

✡✡ $$[u',v'](x) = (-1)^{|u|}u(y)v(z) - (-1)^{|u'||v'|}(-1)^{|v|}v(y)u(z).$$

where $u', v' \in (\Sigma kX)^\vee$ are shifted versions.

Rmk: Note $\langle -,-\rangle$ well-def. since ∂ quadratic and $u\otimes v - (-1)^{|u||v|} v\otimes u$ anti-symmetric.

② 2.4.7 in BEJ: $u, v \in V^\vee$, $x, y \in V$

$$\langle u, \frac{d}{dv}(xy)\rangle = (-1)^{|v|}\langle u'\otimes v' - (-1)^{|u'||v'|} v'\otimes u', xy\rangle.$$

where:

$$\frac{d}{dv}(xy) = v(x)y + (-1)^{|v||x|} x\, v(y), \quad \text{derivation induced by } v \text{ on } V.$$

Pf: Without signs this is straightforward: $u(v(x)y + xv(y))$ is indeed RHS.

Rmk: Def in ① is intuitive: anti-symmetric relation to define $[-,-]$, and extract info. of A by tubing quadratics.

③ Thm: A quadratic model, then $\pi^*(A)$ is graded Lie alg., and:

$$\text{quadmod}_k \underset{\mathcal{L}}{\overset{\pi^*}{\rightleftarrows}} \text{Lie}_k$$

is an equivalence of cat.

Rmk: Def: L is graded Lie alg. if (over char. 0):

$$\begin{cases} [x,y] + (-1)^{|x||y|}[y,x] = 0. \\ [[x,y],z] = [x,[y,z]] - (-1)^{|x||y|}[y,[x,z]]. \end{cases}$$

Note $|[x,y]| = |x|+|y|$, assuming 0 is in any degree.

L is of finite type if L_n is f.d.